



ElectronicsTutorials

AC CIRCUITS ANALYSIS

For Students, Professionals
and Beyond

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1. AN AC WAVEFORM

By rotating a conductive coil of wire within a magnetic field, we can generate an emf (electro-motive force) which varies sinusoidally with time producing what is commonly called an **Alternating Current**. The abbreviated term “**AC**” is commonly used to refer to an Alternating Current.

The plot, graph, or variation of a generated alternating current or voltage over time is generally known as a **Waveform**. Waveforms are commonly plotted as a function of some variable such as time, degrees, or radians along a horizontal axis.

An alternating waveform is a periodic waveform which alternates between positive and negative values (usually many times per second) cyclically, passing first in one direction, then in the other through a circuit.

Thus an AC waveform varies in both positive and negative magnitude as well as direction in more or less an even manner with respect to time. This means that an AC Waveform is a bi-directional time-dependent waveform.

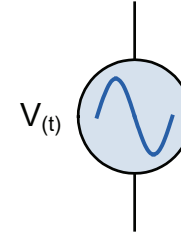
AC waveforms periodically reverse direction with the most common being the sinusoidal waveform

By convention, alternating currents are called AC currents and alternating voltages are called AC voltages. There are many different types and shapes of waveforms but the most fundamental is called the sine wave (or sinusoid).

The sine wave or sinusoidal AC waveform is the most common voltage and current waveform shape that supplies energy to the wall socket outlets in your home. The advantage of the alternating waveform for electric power is that it can be stepped up or stepped down in potential easily for transmission and utilisation.

Alternating Current (AC) circuit theory focuses on the mathematical analysis of electrical circuits in which the voltages and currents applied to them varies periodically with time with the generation of sinusoidal AC voltage explained using mathematical equations. The schematic symbol used to denote an AC voltage source is given as:

FIGURE 1. AN AC VOLTAGE SOURCE



In AC circuits the voltages and currents supplied by a source are not constant but are defined by sinusoidal functions. This means that their electrical values change over time with reference to the period of the sinusoidal waveform.

Thus in AC circuits the voltages and currents are defined by the following sinusoidal functions.

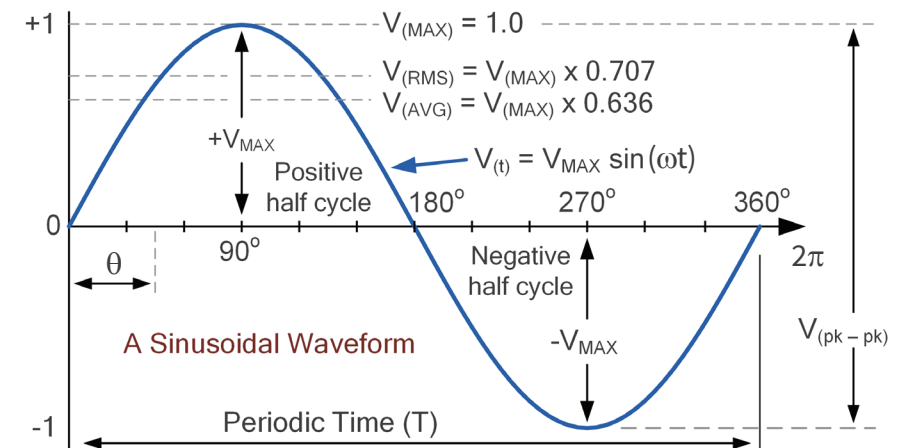
$$\text{Voltage, } v_{(t)} = v_{\text{MAX}} \sin(\theta) = v_{\text{MAX}} \sin(\omega t)$$

$$\text{Current, } i_{(t)} = i_{\text{MAX}} \sin(\theta) = i_{\text{MAX}} \sin(\omega t)$$

Where the maximum value, v_{MAX} is called the magnitude or amplitude of the waveform. Since AC waveforms vary with time, they are designated by lowercase letters $v(t)$ for voltage, and $i(t)$ for current instead of the uppercase letters V and I used for steady state DC values. Note that the subscript (t) is the representation of time.

A typical sinusoidal waveform of voltage as a function of time is shown in Figure 2.

FIGURE 2. A SINUSOIDAL WAVEFORM



The waveform of Figure 2. starts at zero, increases to a positive maximum (also called the peak), and then decreases to zero, changes polarity, increases to a negative maximum, then returns again back to zero.

One complete variation between the same points on the waveform is commonly referred to as one complete *cycle*. Since the waveform repeats itself at regular intervals over time, it is therefore known as a *periodic waveform*. Thus the periodic or AC waveform is the resulting product of a rotating electrical generator.

The type and shape of an AC waveform depends upon the generator or device producing them, but all AC waveforms consist of a zero voltage line that divides the waveform into two symmetrical halves. The main characteristics of an AC Waveform are defined as:

1. **The Amplitude, (A)** also referred to as the maximum value or peak value is the maximum voltage or current reached by the waveform.
2. **The Period, (T)** is the length of time in seconds that the waveform takes to repeat itself from start to finish. This can also be called the Periodic Time of the waveform for sine waves, or the Pulse Width for square waves.
3. **The Frequency, (f)** is the number of times the waveform repeats itself within a one second time period. Frequency is the reciprocal of the time period, ($f = 1/T$) with the unit of frequency being the Hertz, (Hz). The angular frequency (ω) is defined to be $\omega = 2\pi f$
4. **The Phase Angle, (Φ)** is the phase difference between two sinusoidal waveforms of the same frequency.

2. IMPEDANCE OF AN AC CIRCUIT

All electrical circuits contain *Resistors* (R), *Inductors* (L), *Capacitors* (C), or combinations of these three passive components. The operation and performance of an AC circuit will be different for each component used causing certain circuit properties to exist. But first let us look at an AC circuit where only one of these R, L, or C components is connected across a sinusoidal voltage supply.

Any ideal basic circuit element such as a resistor can be described mathematically in terms of its voltage and current, and we saw previously that the voltage across a pure ohmic resistor is linearly proportional to the current flowing through it as defined by Ohm's Law.

For example, if the voltage across a resistor varies sinusoidally with respect to time, as it does in an AC circuit, then the current flowing through the resistor will also vary sinusoidally.

Impedance is the opposition to sinusoidal current flow in an AC circuit being the phasor sum of both reactance and resistance

An AC circuit consisting of reactance, either inductive or capacitive, and a resistance will have an equivalent AC opposition to current flow known as **Impedance, Z** (measured in Ohm's). Thus in an AC circuit, impedance represents the opposition to the flow of sinusoidal current.

Impedance is not the sum or addition of reactance, X and resistance, R but is the result of a phasor sum. Note that although impedance represents the ratio of two phasors, it is not a phasor in itself, because it does not correspond to a sinusoidal varying quantity.

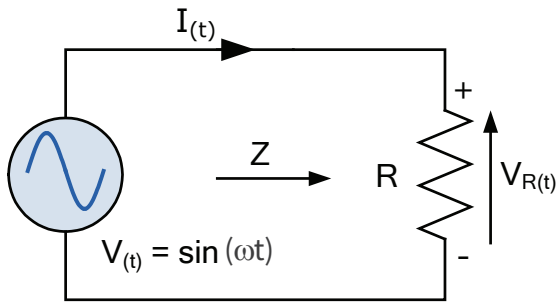
Impedance, which is given the letter Z, can be presented as a complex number. A pure ohmic resistance has a complex number consisting only of a real part, being the actual AC resistance value, (R) and a zero imaginary part, giving ($Z = R \pm j0$).

Likewise, for a component with pure reactance, (X) there is zero real part and only an imaginary part, giving ($Z = 0 \pm jX$). As we will see later in this AC circuits ebook, a pure reactance has a phase angle which is $\pm 90^\circ$ "out-of-phase".

3. RESISTANCE IN AN AC CIRCUIT

The most basic of all AC circuits is that of a resistive circuit in which a pure resistor (a pure resistor is one which exhibits only electrical resistance and zero reactance) is connected directly to an AC supply as shown in Figure 3.

FIGURE 3. A RESISTIVE AC CIRCUIT



Resistors are “passive” devices, that is they do not produce or consume any electrical energy, but convert electrical energy into heat or light.

The current $I_{(t)}$ flowing through a resistor (R) is directly proportional to the instantaneous voltage, and therefore inversely proportional to the resistive value of the resistor.

The supply voltage ($V_{(t)}$) is varying sinusoidally so the resistance has a constant resistive value. It is important to note, that when used in AC circuits, a resistance will always have the same resistive value no matter the supply frequency.

In other words, supply frequency has not effect on resistance. Because of this we can use both Ohms Law and Kirchhoff’s circuit laws, as well as simple circuit rules for calculating the voltage, current, impedance and power of an AC circuit as we would in DC circuit analysis.

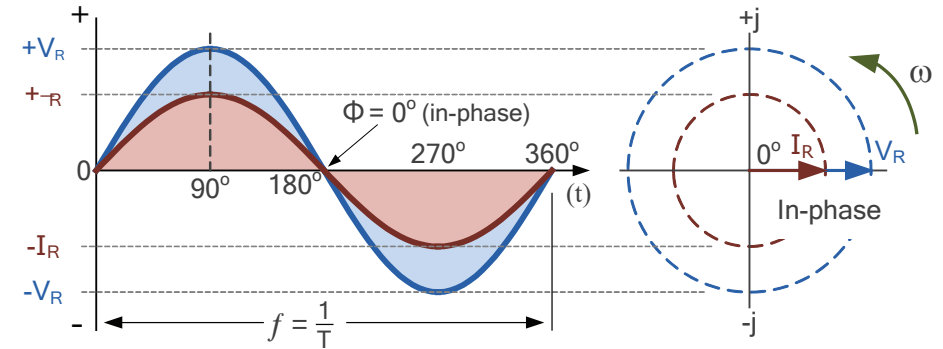
Then the impedance of a resistor is defined as:

$$Z = R = \frac{V_R}{I_R} = (R + j0) \text{ Ohms}$$

The current flowing around a purely resistive circuit will also be sinusoidal, and in-phase with the applied voltage. The term “in-phase” means that both the voltage and current waveforms will reach their maximum (peak) value and cross through the zero axis at exactly the same moment in time along the graph’s horizontal axis.

In other words, the phase difference, or phase shift between the voltage and current waveforms in a resistive circuit is zero (0°).

FIGURE 4. VOLTAGE-CURRENT RELATIONSHIP



We can see in Figure 4. that at any point along the horizontal axis that the instantaneous voltage and current are “in-phase” because the current and the voltage reach their maximum values at the same time, that is their phase angle θ is 0° . Then these instantaneous values of voltage and current can be compared to give the ohmic value of the resistance simply by using Ohm’s law as $I = V/R$.

$$i_{MAX} \sin(\omega t) = \frac{V_{MAX} \sin(\omega t)}{R}$$

$$\therefore i_{MAX} = \frac{V_{MAX}}{R} \text{ Amps, or } i_{RMS} = \frac{V_{RMS}}{R} \text{ Amps}$$

Note that when working with AC alternating voltages and currents, unless otherwise stated, it is usual to use only “RMS” values to avoid confusion. The RMS or “Root Mean Squared” value of an AC waveform is the effective or equivalent DC value for an AC waveform.

The instantaneous power (p), in a resistive circuit at any instant in time can be found by multiplying the instantaneous voltage and current at that instant. Since for resistors in AC circuits, the phase angle ϕ between the voltage and the current is zero (in-phase), then the power factor of the circuit is given as $\cos 0^\circ = 1.0$.

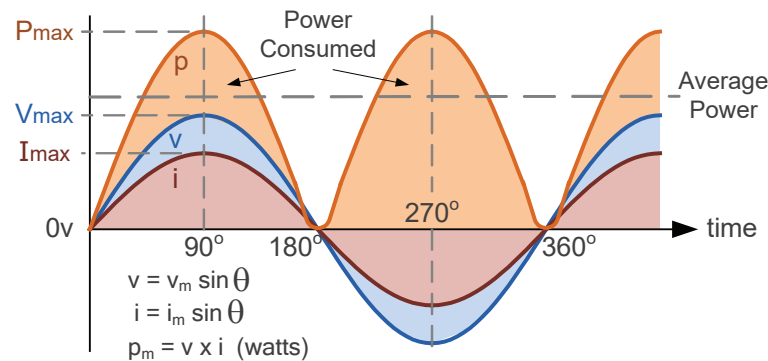
So for a purely resistive circuit, the power consumed is simply given as: $p = v \times i$, the same as for Ohm's Law. Thus the power dissipated in a purely resistive load fed from an AC rms supply is the same as that for a resistor connected to a DC supply and is given as:

$$P = V_{\text{RMS}} \times I_{\text{RMS}} = I_{\text{RMS}}^2 \times R = \frac{V_{\text{RMS}}^2}{R} \text{ Watts}$$

Where: P is the power in Watts, V_{RMS} is the voltage, I_{RMS} is the current, and R is the resistance in Ohm's

The power in a purely resistive AC circuit will be a series of positive pulses because when the voltage and current are both in their positive half of the cycle the resultant power is positive. When the voltage and current are both negative, the product of the two negative values also gives a positive power pulse as shown in Figure 5.

FIGURE 5. POWER IN A PURELY RESISTIVE AC CIRCUIT



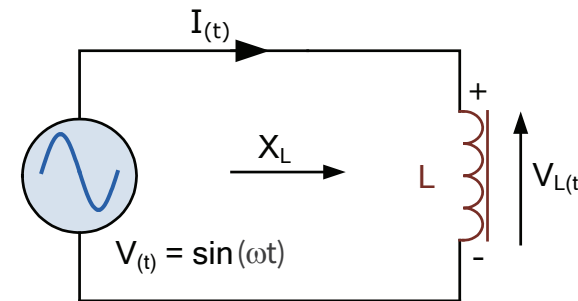
So we can see that the average power dissipated by a pure resistor in an AC circuit is exactly the same as the electrical power dissipated by a pure resistor in an equivalent DC circuit. That is $P_{\text{AC}} = P_{\text{DC}}$.

4. INDUCTANCE IN AN AC CIRCUIT

When an alternating or AC voltage is applied across an inductor (coil) the flow of current through it behaves very differently to that of an applied DC voltage. The effect of a sinusoidal supply produces a phase difference between the voltage and the current waveforms. In an AC circuit, the opposition to current flow through an inductor's coil windings not only depends upon the inductance of the coil but also the frequency of the AC supply.

The opposition to current flowing through a coil in an AC circuit is determined by the reactance (X), of the inductive coil. As the component we are interested in is an inductor, the reactance (X) of an inductor is therefore called "**Inductive Reactance**". In other words, an inductor's electrical resistance when used in an AC circuit is commonly known as: Inductive Reactance.

FIGURE 6. AN INDUCTIVE CIRCUIT



Then inductive reactance is the quantity that represents the opposition a coil's inductance presents to current flow in an AC circuit.

While a pure inductor has no electrical resistance. Inductive reactance, which is given the symbol X_L , is the property in an AC circuit which opposes the change

of the current due to the back emf induced within the coil.

In an AC inductive circuit, this inductive reactance value, X_L is given as: $2\pi fL$ or $j\omega L$. Thus:

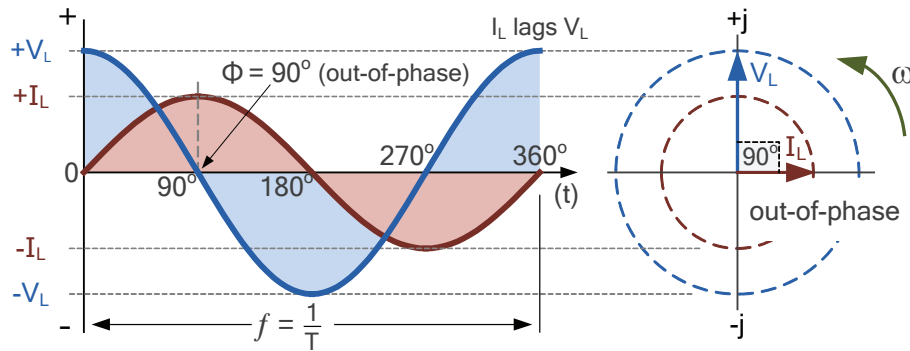
$$Z = X_L = 2\pi fL = \omega L = \frac{V_L}{I_L} = (0 + jX_L) \text{ Ohms}$$

Where: X_L = Inductive Reactance (ohms), f = frequency (Hertz), L = inductance (Henries)

In a purely inductive AC circuit, as the supply voltage increases and decreases with the frequency, the self-induced back emf of the coil also increases and decreases with respect to this change. This self-induced emf is therefore directly proportional to the “rate of change of the current” through the coil and is at its greatest potential when the supply voltage crosses over from its positive half cycle to its negative half cycle or vice versa at points, 0° and 180° along the sine wave.

Consequently, the minimum rate of change of the self-induced voltage occurs when the AC sine wave crosses over at its maximum or minimum peak voltage level. At these positions in the cycle the maximum or minimum currents are flowing through the inductor circuit as shown in Figure 7.

FIGURE 7. VOLTAGE-CURRENT PHASE RELATIONSHIP



The voltage and current waveforms show that for a purely inductive circuit, the coils current lags the voltage by 90° (I_L “lags” V_L by 90°). Likewise, we can also say that the voltage leads the current by 90° (V_L “leads” I_L by 90°). Thus the voltage and current are both “out-of-phase” from each other, therefore $\Phi = +90^\circ$ (or $+\pi/2$ rads).

Note that for a series AC circuit, the current is the same throughout the circuit. Therefore, the current is used as the reference quantity when discussing the phasor relationships between the voltage and current in the circuit.

There are many different ways to remember the phase relationship between the voltage and current flowing through a purely inductive circuit. But one very simple and easy way to remember is to use the mnemonic expression “ELI”.

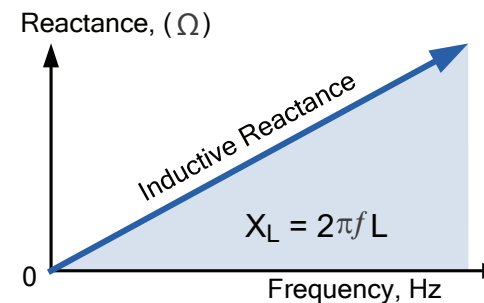
ELI stands for Electromotive force first in an AC inductance, L before the current I. In other words, voltage before the current in an inductor, E, L, I, becomes equal to “ELI”, and whichever phase angle the voltage starts at, this expression always holds true for a pure inductor circuit.

5. EFFECT OF FREQUENCY ON INDUCTIVE REACTANCE

We saw previously that inductive reactance (X_L) is equal to $2\pi fL$ or $j\omega L$. So it can be seen that if either the frequency (f) or the coils inductance (L) is increased, the overall inductive reactance value of the coil would also increase. So as the applied frequency increases towards infinity the inductive reactance (X_L) would also increase towards infinity until it was high enough to act like an open circuit.

Likewise, as the applied frequency reduces to zero or DC, the inductive reactance would also decrease to zero, acting like a short circuit. This means then that inductive reactance is “proportional” to frequency. In other words, inductive reactance increases with frequency resulting in X_L being small at low frequencies and X_L being high at high frequencies as shown in Figure 8.

FIGURE 8. INDUCTIVE REACTANCE AGAINST APPLIED FREQUENCY

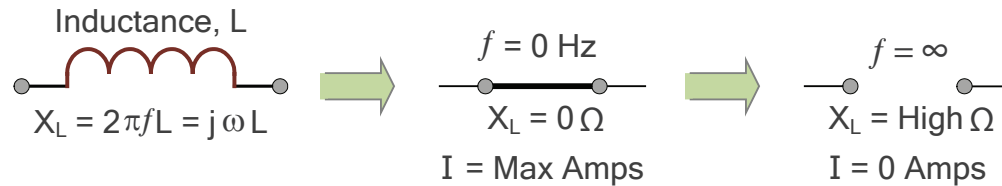


The slope of the graph shows that the inductive reactance of an inductor increases as the applied frequency increases.

Therefore, we can say that a coils inductive reactance is proportional to frequency ($X_L \propto f$) as the back emf generated in the inductor is equal to its inductance multiplied by the rate of change of current through the inductor.

We can present the effect of very low and very high frequencies on the reactance of a purely inductive AC circuit as shown in Figure 9.

FIGURE 9. EFFECT OF FREQUENCY ON INDUCTIVE REACTANCE

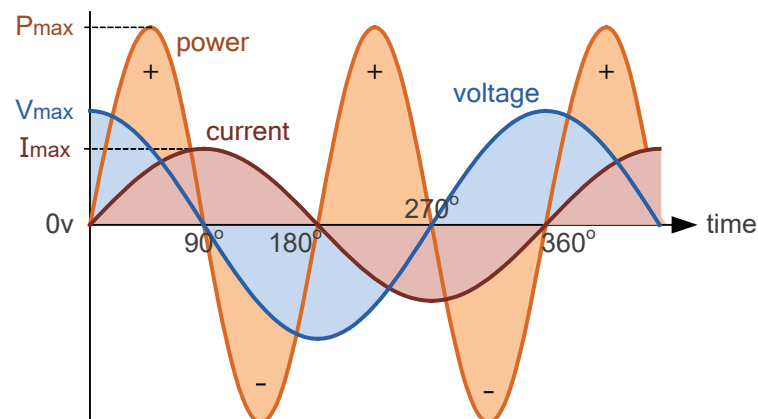


That is, at steady state DC voltages, an inductive coil acts as a short circuit (X_L reduces to 0) and high AC frequencies, an inductive coil acts as an open circuit (X_L approaches infinity).

The instantaneous power (p), in a purely inductive circuit at any instant in time can be found by multiplying the instantaneous voltage and current at that instant. Since the voltage and current waveforms for a purely inductive circuit are always “out-of-phase” from each other by 90° (or $\pi/2$ rads), the phase angle ϕ between the voltage and current waveforms results in a power factor of $\cos(90^\circ) = 0$.

So for a purely inductive circuit, the power consumed is zero as shown in Figure 10.

FIGURE 10. POWER IN A PURELY INDUCTIVE AC CIRCUIT



In the first quarter cycle (0 to 90°), both v and i are positive in value. The resulting v times i power product is therefore positive in this quarter cycle because the product of two positive values is positive. In the next quarter cycle (90 to 180°), the current is still positive, but the voltage is now negative.

The resulting v times i power waveform is therefore negative as the product of the two values is negative. This positive, negative sequence is repeated every half cycle along the waveform thereafter with two positive values and two negative values being created every full cycle.

The result is that the addition of the two positive values with the exact and opposite negative values results in the average power during one full cycle being zero as they cancel each other out. Thus, a purely inductive circuit dissipates zero power.

In reality during the first quarter cycle, the magnetic field created around the coil by the current flowing through it stores electrical energy. In the second quarter cycle, the magnetic field collapses delivering all its stored energy back to the circuit. This sequence of storing energy and delivering energy is repeated every half cycle at a rate determined by the supply frequency.

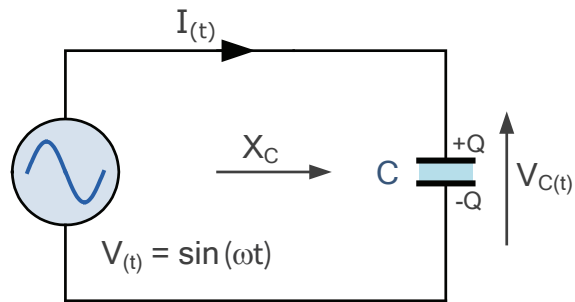
Therefore, an inductor coil returns as much energy as it receives, so the average power consumed in a purely inductive circuit (90° phase angle) is zero. Note then that inductors, coils, windings and solenoids “store” electrical energy in the form of a magnetic field.

6. CAPACITANCE IN AN AC CIRCUIT

A capacitor is a device which stores electric charge and consists basically of two parallel conductive plates separated by an insulating material called a dielectric which is used to maintain a physical separation of the two plates.

When an alternating sinusoidal voltage is applied to the plates of a capacitor, the capacitor is first charged in one direction and then again in the opposite direction, changing the voltage polarity on its plates at a rate determined by the frequency of the AC supply voltage.

FIGURE 11. A CAPACITIVE CIRCUIT



This instantaneous change in voltage across the capacitor is opposed by the fact that it takes a certain amount of time to deposit (or release) the charge on the plates.

Capacitors oppose these changes in sinusoidal voltage with the flow of electrons through the capacitor being directly proportional to the rate of

voltage change across its plates during the process of charging and discharging.

The opposition to current flowing through a capacitor in an AC circuit is determined by the reactance (X), (the same as for an inductive coil). As the component we are interested in is a capacitor, the reactance of a capacitor is therefore called “**Capacitive Reactance**”, (X_C) measured in ohms.

For an AC capacitance circuit, this capacitive reactance value, X_C is equal to $1/(2\pi fC)$ or $1/(j\omega C)$. Thus:

$$Z = X_C = \frac{1}{2\pi fC} = \frac{1}{\omega C} = \frac{V_C}{I_C} = (0 + jX_C) \text{ Ohms}$$

Where: X_C = Capacitive Reactance (ohms), f = frequency (Hertz), C = capacitance (Farads)

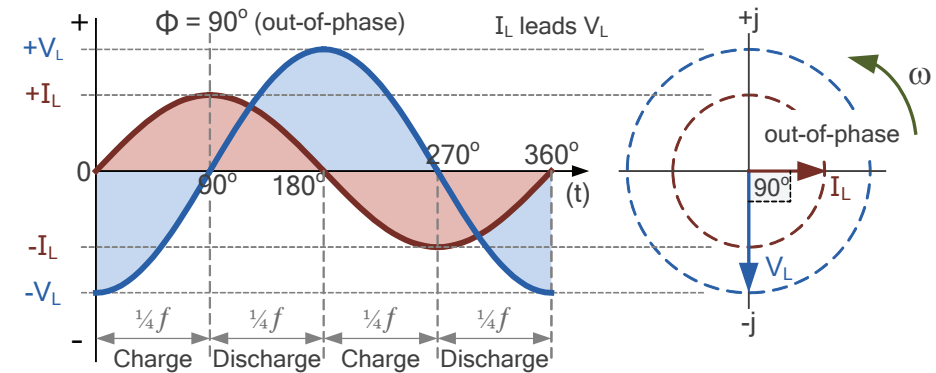
In a purely capacitive AC circuit, as the supply voltage increases and decreases with the frequency, so too does the charge (Q) as the charge on the capacitor is directly proportional to the voltage across it. The resulting waveform given for the capacitor charge will therefore be in-phase with the applied voltage.

As current is the rate of change of charge, the charge on the capacitors plates will be at its maximum when the current is at its maximum, and minimum when the current is at its minimum. Since the greatest amount of current would flow through the capacitor when the voltage change is at its most rapid.

Thus the rate of change of charge is maximum when the waveform passes through the zero axis as the voltage changes polarity.

Then the current through the capacitor will therefore be at its maximum values at these 0° , 180° , 360° points as shown in Figure 12.

FIGURE 12. VOLTAGE-CURRENT PHASE RELATIONSHIP



Then we can see that for a purely capacitive circuit, the current leads the voltage by 90° (I_L “leads” V_L by 90°). Likewise, we can also say that the voltage lags the current by 90° (V_L “lags” I_L by 90°). Thus the voltage and current are both “out-of-phase” from each other, therefore $\Phi = -90^\circ$ (or $-\pi/2$ rads).

To maintain consistency, the AC current will be considered as the reference quantity when discussing phasor relationships between the voltage and current in a capacitive circuit.

There are also different ways to remember the phase relationship between the voltage and current in a purely capacitive circuit, with the mnemonic expression “ICE” being the easiest to remember. ICE stands for current (I) first in an AC capacitance, (C) before Electromotive force (E).

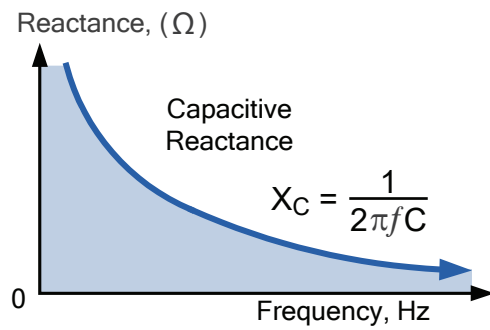
In other words, current before the voltage in a capacitor, I, C, E, equals “ICE”, and whichever phase angle the voltage starts at, this expression always holds true for a pure AC capacitance circuit.

7. EFFECT OF FREQUENCY ON CAPACITIVE REACTANCE

Since capacitors charge and discharge in proportion to the rate of voltage change across them, the faster the voltage changes the more current will flow. Likewise, the slower the voltage changes the less current will flow.

That is, the reactance of an AC capacitor is “inversely proportional” to the applied frequency. As inductive reactance (Ohms) decreases with frequency resulting in X_L being high at low frequencies and X_L being small at high frequencies as shown in Figure 13.

FIGURE 13. CAPACITIVE REACTANCE AGAINST APPLIED FREQUENCY

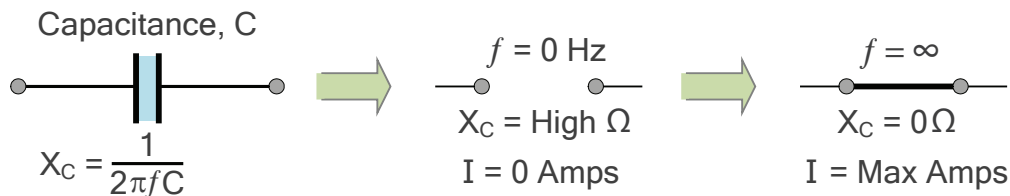


The slope of the graph shows that the capacitive reactance of a capacitor decreases as the applied frequency increases.

Therefore, we can say that a capacitors reactance is inversely proportional to frequency ($X_C \propto 1/f$) for any given capacitance value as the voltage across the capacitor is limited by the rate at which electric charge can be stored, or released during the charging and discharging phases.

We can present the effect of very low and very high frequencies on the reactance of a purely inductive AC circuit as shown in Figure 14.

FIGURE 14. EFFECT OF FREQUENCY ON CAPACITIVE REACTANCE

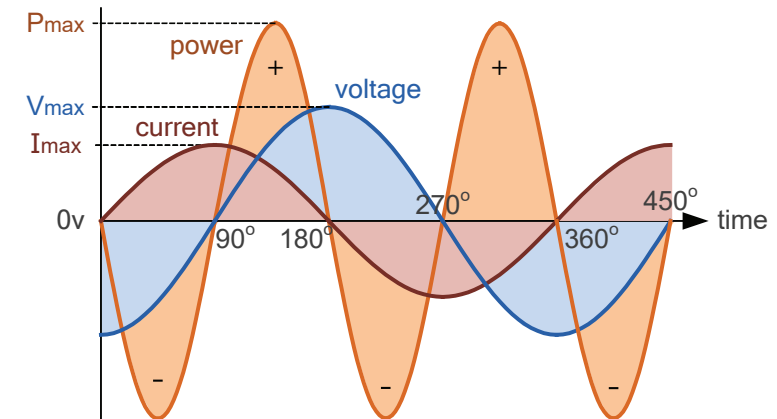


Thus at steady state DC voltages, a capacitor acts as an open circuit (X_C approaches infinity) and high AC frequencies, a capacitor acts as a short circuit (X_C reduces to zero).

Again, the instantaneous power (p), in a purely capacitive circuit at any instant in time can be found by multiplying the instantaneous voltage and current at that instant. Since the voltage and current waveforms for a purely capacitive circuit are always “out-of-phase” from each other by -90° (or $-\pi/2$ rads), the average power consumed will be zero.

So for a purely capacitive circuit, the power consumed is given as: $p = v \times i \times 0 = 0$ watts as the positive and negative half-cycles cancel each other out as shown in Figure 15.

FIGURE 15. POWER IN A PURELY CAPACITIVE AC CIRCUIT



As with the previous inductor, the capacitor is storing energy and delivering energy back into the connected circuit every half cycle at a rate determined by the supply frequency. Therefore, it returns as much energy back into the connected circuit as it receives since i leads v by 90° .

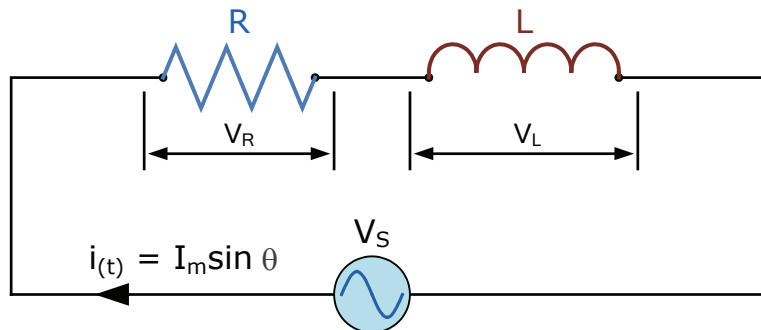
Note that capacitors “store” energy in their electric field at an amount determined by the energy density per unit volume stored in the space between the plates. Thus the amount of energy density a capacitor has is determined by the size of their plates (A), and the distance they are apart (d).

8. SERIES RL CIRCUIT ANALYSIS

A pure inductor maybe one which possesses inductance only, but in the real world pure inductive devices do not exist as every inductor consists of a coil of copper wire which has therefore a resistive value. Thus an inductor must possess a value of resistance (R) in addition to its inductance (L).

For an inductor, coil, solenoid, relay, or any winding for that matter it must consist of a resistive value connected in series with an inductive value to form what is commonly referred to as a *Resistor-Inductor* (RL) series circuit as shown in Figure 16.

FIGURE 16. A SIMPLE RL SERIES CIRCUIT



In an RL circuit there exists a combination of resistance and inductive reactance that opposes the current flow. The total opposition to the flow of current around an AC circuit depends on the values of both resistance and reactance, as well as the phase angle between them. This total opposition to current flow in an AC circuit is called Impedance, Z, measured in ohms.

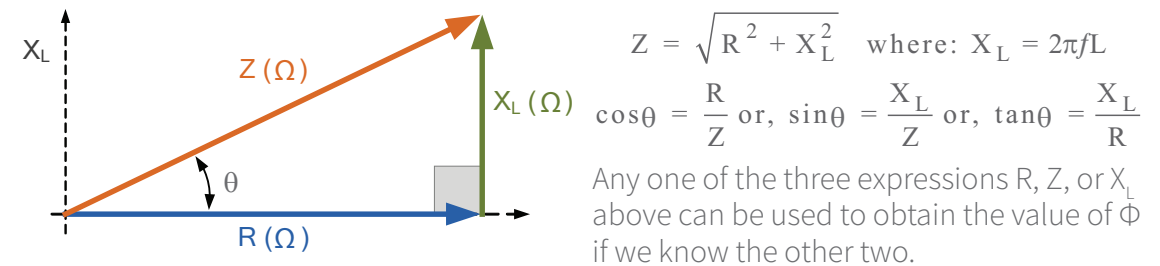
The total impedance of an RL circuit can be calculated by adding the resistance and reactance of the circuit vectorially as two phasors by simply applying Pythagorean theorem. The mathematical expression to define impedance as a phasor sum of R and X_L in an AC circuit is given as:

$$Z^2 = R^2 + X^2 \quad \text{or} \quad Z = \sqrt{R^2 + X^2}$$

Where: Z = impedance (Ω), R = resistance (Ω), and X = reactance (Ω).

What is commonly called an “impedance triangle” can be used to find the relationship between impedance, resistance, and reactance in a series RL circuit is shown in Figure 17.

FIGURE 17. RIGHT HAND TRIANGLE RELATIONSHIP OF Z, R AND X_L



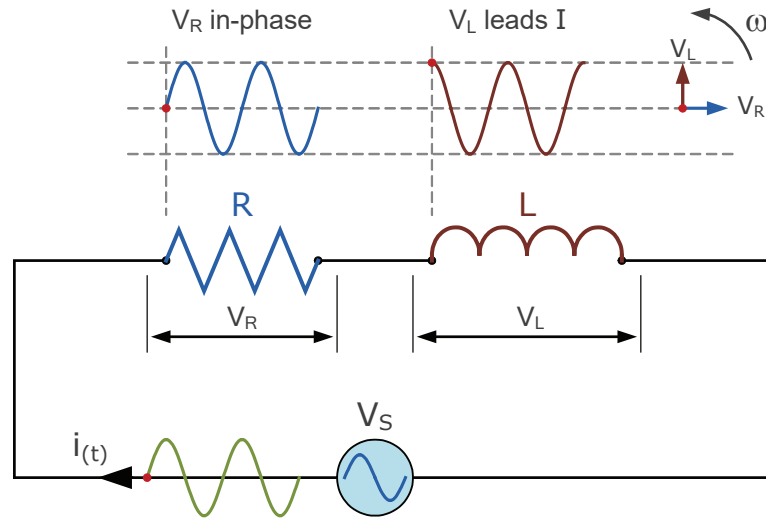
So for example, if $R = 60\Omega$ and $X_L = 80\Omega$, then $Z = 100\Omega$ and the corresponding phase angle (Φ) between the two would be 53.13° .

In a series RL circuit, the current (I) through the resistor and inductor is the same, as it is in any series circuit. Then the flow of this circuit current will produce a V_R voltage drop across the resistor, due to its resistive value.

Similarly, the inductor will produce a V_L voltage drop due to its inductive reactance, X_L . However, the sum of the voltage drops across the resistor and inductor, do not directly add algebraically together to equal the value of the applied voltage.

The voltage drop across the resistor will be “in-phase” with the current through it, thus its phase angle, $\Phi = 0$. However, as we have seen previously, the voltage dropped across the inductor “leads” the current by 90° . Thus $\Phi = +90^\circ$. Then the total applied voltage, V_s will be equal to the phasor sum of V_R and V_L as shown in Figure 18.

FIGURE 18. SERIES R_L CIRCUIT PHASOR DIFFERENCE



We can again use the same Pythagorean theorem and mathematical equations as for the impedance triangle above to find the relationships between the voltage drops in the circuit and the applied voltage. That is, a voltage triangle can be created using V_R , V_L and V_S . Any phasor diagram would be drawn with current I as reference (i.e. current I is drawn along the x-axis direction). Thus:

$$(I \times Z)^2 = (I \times R)^2 + (I \times X_L)^2 = V_S^2 = V_R^2 + V_L^2$$

therefore, $V_S = \sqrt{V_R^2 + V_L^2}$

So for example, if $R = 60\Omega$, $X_L = 80\Omega$, and the current $I = 3$ Amperes, then the circuit impedance, $Z = 100\Omega$, resistive voltage drop, $V_R = 180V$, inductive voltage drop, $V_L = 240V$, giving a supply voltage, $V_S = 300V$.

Note then that the algebraic sum of $V_R + V_L = 420V$. Whereas the phasor sum for V_S is $300V$.

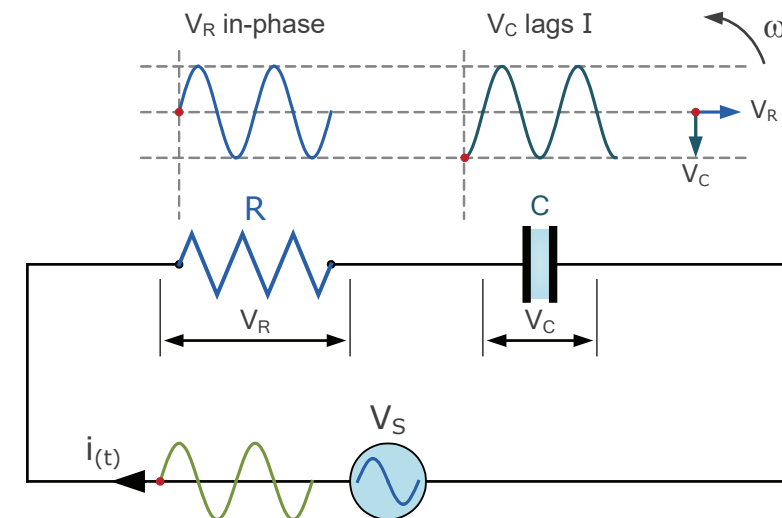
Also note that for a series RL circuit, the current will lag the voltage by a phase angle less than 90° , so the circuit is therefore said to be inductive. As such the power dissipated by this circuit can only occur in the resistive (R) part of the series circuit, and not in the inductor (L). Thus the power consumed by a series RL circuit is exactly equal to the I^2R power loss in the resistor and is not equal to the VI product of the circuit.

9. SERIES RC CIRCUIT ANALYSIS

In an RC series circuit, a pure capacitor and resistor are connected together in series across an AC voltage supply. As before, in a series circuit, the current (i) flowing through the circuit will have the same value through both the resistor and the capacitor as it is common to both components.

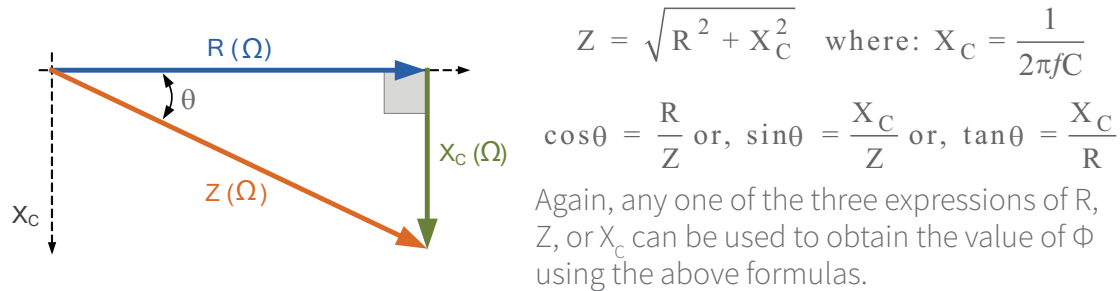
Also the voltage dropped across the resistor is “in-phase” with the current, whereas the voltage across the capacitor lags the current by 90° as shown in Figure 19.

FIGURE 19. SERIES RC CIRCUIT PHASOR DIFFERENCE



Again, as with the previous RL series circuit, the total opposition to current flowing around this RC series circuit is impedance, (Z) which is derived from the combination of R and X_C . Therefore, we can use the same techniques for the RL circuit above to find both the voltage and impedance triangles as shown in Figure 20.

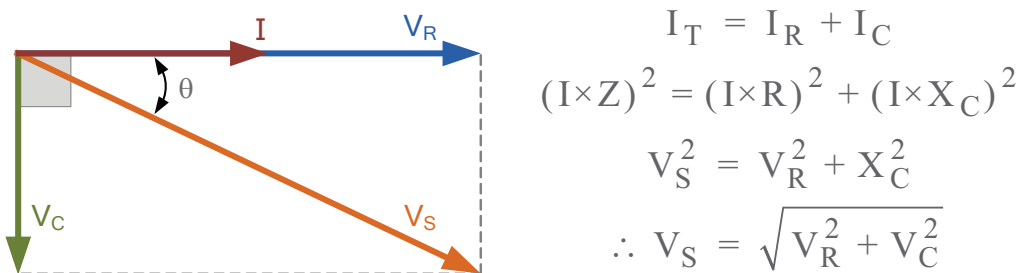
FIGURE 20. RIGHT HAND TRIANGLE RELATIONSHIP OF Z, R AND X_C



As before, we can use the same Pythagorean theorem and mathematical equations as for the impedance triangle to find the relationships between the KVL voltage drops around the series RC circuit. A voltage triangle can be created using V_R , V_C and V_S .

Any phasor diagram would be drawn with current, i as reference (i.e. current (i) is drawn along the x-axis direction). Thus:

FIGURE 21. VOLTAGE PHASOR DIAGRAM



Again, while the algebraic sum of the individual voltage drops is more than the applied voltage V_S , the vector sum of V_R and V_C is equal to V_S due to the phase difference.

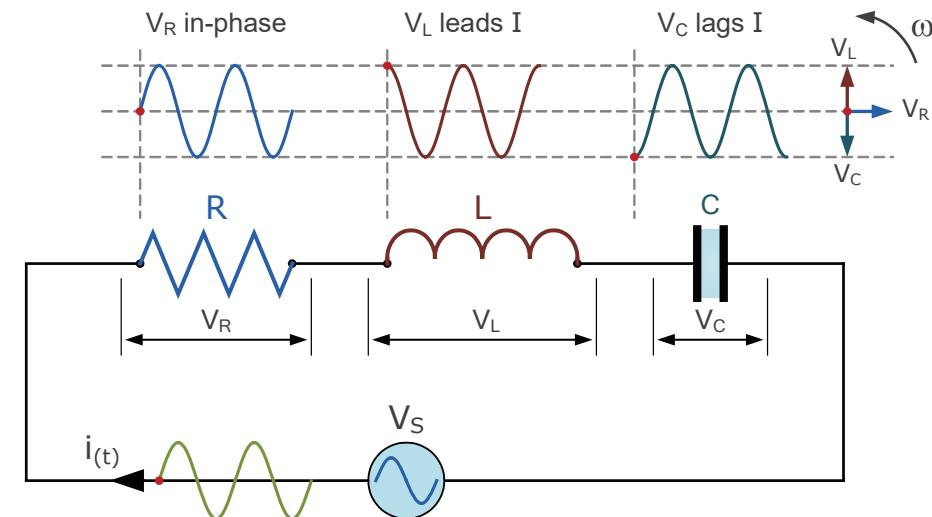
Note that for an RC series circuit, the current leads the voltage by a phase angle less than 90° , so the circuit is said to be capacitive. As such the power dissipated by this circuit can only occur in the resistance (R) and not the capacitor (C). Thus the power consumed by an RC series circuit is equal to the I^2R power loss in the resistor and is not equal to the VI product of the circuit.

10. SERIES RLC CIRCUIT ANALYSIS

As its name would suggest, an RLC series circuit consists of a resistor, (R) an inductor, (L) and a capacitor, (C) all connected together in a series combination across an AC supply.

We have seen previously that in a pure ohmic resistor the voltage waveform is “in-phase” with the current. In a pure inductance the voltage waveform “leads” the current by 90° , giving us the expression of “ELI”. In a pure capacitance the voltage waveform “lags” the current by 90° , giving us the expression of “ICE”. Consider the RLC circuit of Figure 22.

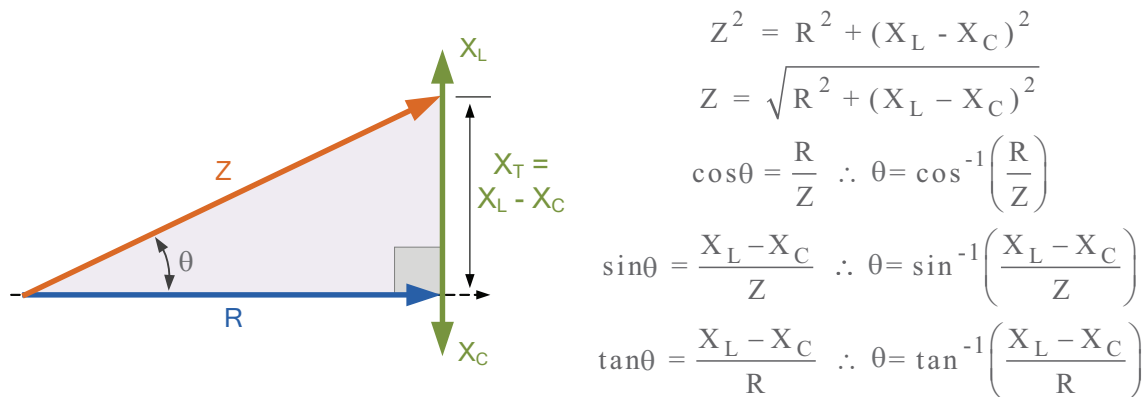
FIGURE 22. A SERIES RLC CIRCUIT



The phase difference, Φ between the voltage and current depends upon the reactive value of the components being used and hopefully by now we know that reactance, (X) is zero if the circuit element is resistive, positive if the circuit element is inductive and negative if it is capacitive.

As the three vector voltages are effectively “out-of-phase” with each other, X_L , X_C and R must also be “out-of-phase” with each other. Thus the relationship between R , X_L and X_C will be the vector sum of these three components. Then the mathematical representation of an RLC circuit to give us the overall impedance, Z is shown in Figure 23.

FIGURE 23. RLC IMPEDANCE TRIANGLE



Note that the ohmic difference between X_L and X_C is squared so the order by which these two quantities are subtracted from each other, $X_L - X_C$, or $X_C - X_L$ will have no effect on the final answer.

Also note that if $X_C > X_L$ then the phase angle Φ of the circuit will be lagging, so the circuit is inductive. If $X_L > X_C$ then the phase angle will be leading, so the circuit is capacitive.

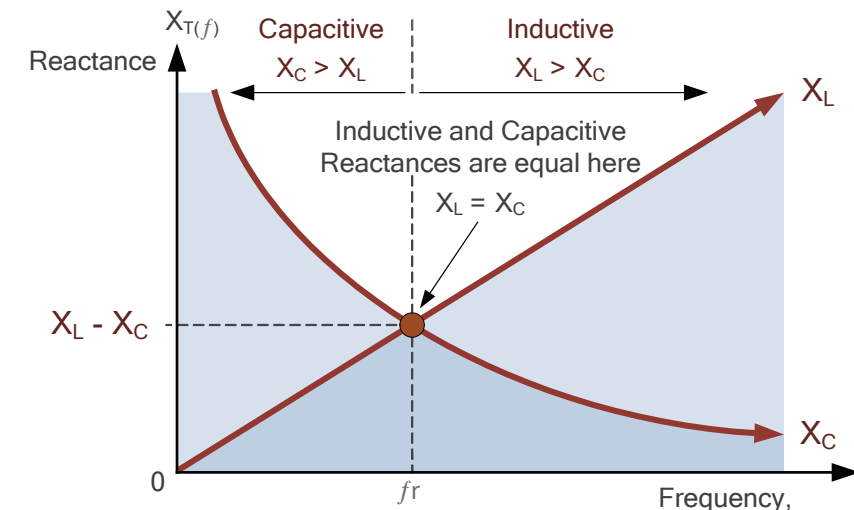
11. RESONANCE IN A SERIES RLC CIRCUIT

In a series RLC circuit with a varying frequency, there becomes a frequency point where the inductive reactance of the inductor becomes equal in value to the capacitive reactance of the capacitor. In other words, $X_L = X_C$. At this point the total circuit reactance of, $X_T = X_L - X_C$ reduces to zero and the circuit impedance becomes totally resistive.

The point at which this occurs is called the *Resonant Frequency* point, (f_r) of the circuit. As in this example we are analysing a series RLC circuit this resonance frequency produces what is known as **Series Resonance**. Then as both inductive reactance and capacitive reactance are dependent on the supply frequency, it is possible to bring a series RLC circuit into resonance simply by adjusting the frequency of the applied voltage supply.

We can demonstrate this by placing the graphs of Figure 8. and Figure 13. on top of each other so that both lines are on the same x- axes. Thus the point of intersection will give us the series resonance frequency point as shown in Figure 24.

FIGURE 24 RESONANCE FREQUENCY POINT



The frequency point at which the two reactance's are exact, opposite and equal so cancelling each other out occurs where the two reactance curves cross each other as shown on the graph of Figure 24.

For the previous series RLC circuit, when at resonance the net reactance of the circuit will be zero as: $X_T = X_L - X_C = 0$. In complex form, the resonant frequency is the frequency at which the total impedance of a series RLC circuit becomes purely "real", that is no imaginary impedance exists.

So at this point the total impedance of the series RLC circuit simply becomes equal to the circuit resistance ($Z = R$) and therefore at its minimum value. Then for a series RLC resonant circuit, the resonant frequency, (f_r) point can be calculated as follows.

$$X_L = X_C \Rightarrow 2\pi fL = \frac{1}{2\pi fC}$$
$$\therefore f_r = \frac{1}{2\pi\sqrt{LC}} \text{ (Hertz)}$$

As the circuit impedance is minimum at resonance, the current flowing through the circuit will be at its maximum value. Thus the applied voltage is in-phase with the current as the phase angle is zero (purely resistive).

This is because the voltage across the inductor (V_L), equals the voltage across the capacitor (V_C). So if $V_L = V_C$ the two voltages cancel each other out since they are 180 degrees out of phase.

Note that in an RLC circuit, the current can either lag or lead the voltage, and the phase angle difference between the current and the voltage can vary between $+90^\circ$ and -90° and the resultant circuit is either inductive or capacitive.

However, under resonant conditions, the voltages created across the capacitor and inductor due to this resonance condition can be many times greater than the supply voltage.

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